

Rutgers University
Master's Comprehensive Examination

Part I

Theory (90 minutes)

December 5, 1987

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded.
Open book and open notes.

1. Let $X_1, X_2, \dots, X_n$ be a sample from a population with density $p(x, \theta)$ given by

$$p(x, \theta) = \frac{1}{\sigma} \exp \left\{ \frac{-(x-\mu)}{\sigma} \right\}, \quad x \geq \mu.$$

Here $\theta = (\mu, \sigma)$ with $-\infty < \mu < \infty$, $\sigma > 0$.

- (a) Show that $\min(X_1, X_2, \dots, X_n)$ is sufficient for μ when σ is fixed.
- (b) Find a one-dimensional sufficient statistic for σ when μ is fixed.

2. Suppose that $X_1, X_2, \dots, X_n$ is a sample from a population with density $f(x, \theta) = \theta x^{\theta-1}$, $0 < x < 1$, $\theta > 0$, and let $T(X) = (-1/n) \sum \log X_i$.

- (a) Find $E(T(X))$, $\text{Var}(T(X))$.
- (b) Show that $T(X)$ is an U.M.V.U estimate of $q(\theta) = 1/\theta$.

3. Let $X_1, X_2, \dots, X_n$ and $Y_1, Y_2, \dots, Y_m$ be independent random samples from the distributions $N(\theta_1, \theta_3)$ and $N(\theta_2, \theta_4)$ respectively. Show that the likelihood ratio test for testing $H_0: \theta_3 = \theta_4$, θ_1 and θ_2 unspecified, against $H_1: \theta_3 \neq \theta_4$, θ_1 and θ_2 unspecified, can be based on the random variable

$$F = \frac{\sum_{i=1}^n (X_i - \bar{X})^2 / (n-1)}{\sum_{i=1}^m (Y_i - \bar{Y})^2 / (m-1)}.$$

Give the sampling distribution of the test statistic.

Rutgers University
Master's Comprehensive Examination

Part II

Applied (90 minutes)

December 5, 1987

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded.
Open book and open notes.

1. A researcher measures the pulse rate of four patients before and after a meditation class, with data as follows:

pulse rate	
<u>before</u>	<u>after</u>
72	65
85	75
77	70
65	69

- (a) Is there any evidence that meditation lowers the pulse rate?
- (b) Give a 99% confidence interval for the amount of reduction in the pulse rate during the meditation class.

2. A college professor wishes to know how well a final exam mark can be predicted from a midterm exam mark. The following table gives marks for five students.

midterm exam mark (X):	55	93	88	73	95
final exam mark (Y):	60	85	92	81	85

- (a) Fit a straight line for Y in terms of X and test the hypothesis that the slope of the line equals 1.
- (b) Calculate the correlation coefficient between Y and X and give an approximate 95% confidence interval.

3. In a study of smoking habits, 1000 teachers were asked whether they were nonsmokers, moderate smokers or heavy smokers. The results were broken down by sex. The following table summarizes the number in each category.

	<u>nonsmokers</u>	<u>moderate smokers</u>	<u>heavy smokers</u>	<u>total</u>
men	96	81	103	280
women	335	210	175	720
total	431	291	278	1000

- (a) Is there any evidence that smoking habits differ between men and women?

- (b) Estimate the proportion of nonsmoking females in the teaching population and give the mean and variance of your estimate.

4. A farmer wishes to compare the milk yield of three breeds of cattle. He used three cows from Breed A, four cows from Breed B and one cow from Breed C, with data as follows:

Daily milk yield (in liters)

Breed A:	6, 4, 8
Breed B:	7, 8, 10, 8
Breed C:	10

- (a) Is there any evidence that the daily milk yield differs between the three breeds?
- (b) Give a 95% confidence interval for the difference in daily milk yield between the breed which appears best and the breed which appears worst in the above sample.

5. A Geiger counter is used to count the number of radioactive particles emitted per minute from an object. Two one-gram pieces of rock from a uranium mine are examined with the following counts:

Rock 1:	745
Rock 2:	769

It is known that the number of radioactive particles emitted by a rock follows a Poisson distribution.

- (a) Assuming the two rocks are equally radioactive, estimate the mean number of radioactive particles emitted from each rock per minute.
- (b) Is there any evidence to suggest that the rocks are not equally radioactive?

4. Let X be a random variable with probability density function

$$\frac{1}{\sqrt{8\pi}} \exp \left[-\frac{(x-1)^2}{8} \right], \quad -\infty < x < \infty.$$

Let $Y = X^2 + 2X$. Find the expectation of Y . Also find $E e^X$.

5. Let X be a random variable following the standard normal distribution. Let

$$Y = \frac{1}{\sqrt{2\pi}} \exp (-X^2/2).$$

Find the mean and variance of Y .

Rutgers-The State University
Master's Comprehensive Examination

Part I

Theory

March 28, 1987

- Instructions:
1. Open book and open notes
 2. Do four out of the following five problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded
-

1. Let $T_1, T_2, \dots, T_k$ be independent unbiased estimators of θ with known variances $\sigma_1^2, \dots, \sigma_k^2$, respectively. Find a minimum variance unbiased estimator of θ in the class of all linear functions of $T_1, \dots, T_k$. Prove your answer.
2. Let $X_1, \dots, X_n$ be a random sample from a Poisson distribution with mean λ and let $S_m = \sum_{i=1}^m X_i$, $m < n$.
 - i) Find the conditional distribution of $X_1, \dots, X_n$ given that $S_n = k$.
 - ii) Show that $E(S_m | S_n) = \left(\frac{m}{n}\right) S_n$.
3. Given a random sample of size n from the distribution with density
$$f(x) = (1 + \theta) x^\theta, \text{ for } 0 < x < 1,$$
use the Neyman-Pearson lemma to determine the nature of the best critical region for testing $H_0: \theta = 3$ versus $H_1: \theta = 2$.
4. Let $Y_1, Y_2, \dots, Y_5$ be a random sample from the uniform distribution defined over $(0, \theta)$. Let $X = 5 \text{ Min}(Y_1, Y_2, \dots, Y_5)$. Is X an unbiased estimator of θ ? (Prove or find the bias).
5. Let $X \sim B(n; p)$ and the prior distribution of p is Beta (c, c) .
 - i) Find the Bayes estimator of p if loss is squared error.
 - ii) Find the risk of the estimator in i).
 - iii) Find c so that the risk is a constant.

Rutgers-The State University
Master's Comprehensive Examination

Part II

Applied

March 28, 1987

- Instructions:
1. Open book and open notes
 2. Do four out of the following five problems
 3. Use a separate blue book for this section of the exam
 4. Indicate which questions you wish graded

1. One difficulty in estimating sample size in biological work is that the proportions given by a standard treatment may vary over time. An experimenter has found that his standard treatment has a failure rate lying between $P_1 = 30\%$ and $P_1 = 40\%$. With a new treatment, where failure rate is 20% lower than the standard, what sample size is needed to have 90% power in a two-tailed 5% level test.
2. A nutritionist is interested in comparing the weight gains of guinea pigs fed on four different diets. For each diet 4 animals were selected at random from a group of 16 animals. The diets were then administered for a specified period of time, after which the weight gains were recorded.

		<u>Animals</u>				$\sum_j X_{ij}$	$\sum_j X_{ij}^2$
Diet	1	2	0	-2	0	0	8
	2	0	2	0	-2	0	8
	3	5	5	2	8	20	118
	4	4	0	5	3	12	50
$\sum_i X_{ij}$		11	7	5	9	$\sum_{i,j} X_{ij} = 32$	
$\sum_i X_{ij}^2$		45	29	33	77	$\sum_{i,j} X_{ij}^2 = 184$	

- (a) Test the hypothesis that the mean weight gain is the same for all 4 diets. ($\alpha = .05$.)
- (b) Compare the diet means using a multiple comparison test.
- (c) Using as much of the available information as possible, write a 95% confidence interval for μ_4 , the true mean for diet 4.

3. A friend claims to have E.S.P. (extra-sensory perception). He states that he can predict the result of a toss of a coin with more frequent success than chance would allow.

We believe he is only guessing and decide to test him with 15 flips of a fair coin. Before each flip he "predicts" the results, then we flip the coin and record whether or not he was correct. Assume the trials are independent and identical.

The following questions may be answered using a normal approximation.

- (a) How many correct predictions must he make in 15 trials to persuade us ($\alpha = .05$) that he has these remarkable powers?

H_0 : He is quessing .

H_1 : He has E.S.P.

- (b) If in fact our friend actually can correctly predict the outcome with probability .80, what is the probability that the test you just constructed in part (a) will result in a Type II error?

- (c) Suppose our friend makes 35 correct predictions in 50 trials. Obtain a 95% confidence interval for the true probability that he correctly predicts the outcome of a flip of a coin.

4. A simple linear regression model

$$Y = \beta_0 + \beta_1 X + \epsilon$$

was fit to a set of data. The following results were found.

$$b_1 = 4$$

$$\bar{x} = 4$$

$$\bar{y} = 20$$

Additionally, part of the analysis of variance table is summarized below.

<u>Source</u>	<u>df</u>	<u>ss</u>	<u>ms</u>	<u>F</u>
Total	10	800		
Regression				
Error		288		

- (a) Find the estimated regression line.
- (b) If it is assumed that ϵ is distributed as $N(0, \sigma^2)$ then estimate σ^2 .
- (c) Test the hypothesis that $\beta_1 = 0$. Let the probability of a Type I error = .10.
- (d) Find a 95% confidence interval for the mean value of Y when $X = 4$.
- (e) Find a 95% prediction interval for Y when $X = 4$.
- (f) Find the coefficient of determination. What does this mean?
- (g) Find an estimate of the correlation coefficient, ρ .
- (h) Find a 95% confidence interval for ρ .

5. The following results were obtained from an experiment to determine whether different operators obtained different mean results in routine soil analyses for nitrogen.

day	Operator				
	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>	<u>E</u>
Tuesday	509	512	532	506	509
Wednesday	505	507	542	520	519
Friday	465	472	498	483	475

On each of the 3 days a sample of soil was selected and then divided into five parts. At random these five parts were assigned to the operators to analyze.

- Carry out an appropriate analysis of variance. State any conclusions that you reach.
- State any assumptions that you have made in the analysis in part (a).
- How would your analysis of variance have been different if the experiment had not been blocked?
- Do you think that, in this instance, blocking was worthwhile?